Reinforcement Learning for Ambulance Dispatch and Repositioning

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Indian Institute of Technology Jodhpur — AI Mini Project, Assignment 2

Repository: https://github.com/anujtrivedi06/ambulance-dispatch-repositioning

1. Problem and MDP Formulation

Emergency calls arrive at random locations and times across a city road network. A fleet of K ambulances must answer them, and an agent makes two coupled decisions: **dispatch** (which unit to send to an incoming call) and **repositioning** (where each idle unit should wait). The two conflict. Sending the nearest unit is optimal for the call in hand but can strip cover from a dense district, costing the next call more than it saved this one.

Decisions occur only at irregular event epochs — a call arriving, or a unit becoming free — so the problem is a **semi-Markov decision process** rather than a fixed-step MDP. Transitions carry an elapsed time Δt, and discounting must respect it.

2. State Space (S)

The underlying state is $s = (t, \{(status_k, loc_k, eta_k)\}_{k=1..K}, queue)$: the clock, every unit's status, position and arrival estimate, and the calls still waiting. With |V| = 64 nodes, six unit statuses and K = 6 ambulances, that is on the order of $(64 \times 6)^6 \approx 10^{15}$ configurations before the continuous clock is counted. Exact dynamic programming is impossible and function approximation is mandatory.

The agent therefore observes an **11-dimensional post-decision feature vector**: available-unit fraction; time of day as (sin, cos); expected covered demand; worst-zone coverage; demand-weighted distance to the nearest free unit; four per-zone coverage values; and a bias term. The post-decision state is a deterministic function of (state, action), so the agent can rank every candidate action by one forward pass each, with no sampling.

The feature set was cut from twelve to these after diagnosis on collected data. Three features were exactly redundant (the design matrix came back rank 9 of 12) and the four per-zone coverages were collinear at 0.97 and 1.00, leaving their weights unidentified. Ridge regression hid this: value *prediction* stayed accurate because the collinear directions cancel in-distribution, while the greedy policy degraded, because an action moves along exactly the direction the data could not identify.

3. Action Space (A)

The action space is **discrete**. At a repositioning epoch the agent chooses one of M stations, |A| = M (M = 4 in the main configuration, swept to 8). At a dispatch epoch it chooses an available unit or defers the call to the queue. The learned policy acts on the repositioning choice, which is where the combinatorial difficulty lies; dispatch follows the nearest-available rule, held fixed across every method so that measured differences are attributable to repositioning alone.

4. Reward Function (R)

Reward accrues between decision epochs from the calls *reached* in that interval, which is when an outcome is determined:

$$r_k = \Sigma_{i \in reached[t_k,\, t_{k+1})} \rho(i)$$

with two incentive designs compared. The **binary** design $\rho(i) = 1\{T_i \le \tau\}$, τ = 8 min, is the operational EMS standard. The **shaped** design $\rho(i) = -min(T_i, C)/10$, C = 40 min, is continuous in response time.

Because epochs are irregularly spaced, discounting is continuous-time: $\gamma_k = exp(-\beta\Delta t)$ with β = 1/240 per minute. A flat per-decision discount would value a decision followed by a quiet night identically to one followed by a busy minute.

This choice dominated every other design decision in the project. The binary indicator cannot distinguish a 3-minute response from a 7-minute one, so it starves learning of gradient over most of the space it cares about. Switching to the shaped reward is worth +0.052 coverage for the linear agent and +0.148 for the neural one — larger than the difference between linear and neural function classes.

5. Environment and Simulation Dynamics

The environment is a purpose-built discrete-event simulator (no third-party RL environment was used). Calls follow a non-homogeneous Poisson process sampled by thinning against a 24-hour demand profile, with locations drawn from a mixture of uniform background and Gaussian hotspots. Travel follows shortest paths over a weighted road graph with multiplicative noise; on-scene durations are lognormal; 75% of calls require transport to hospital plus a 15-minute handover. The main configuration is an 8×8 grid city, 4 stations, 6 ambulances and 4 calls/hour, which holds fleet utilisation near 65% — loaded enough that poor repositioning costs response time, slack enough that good repositioning has somewhere to put units.

Every policy is evaluated on **identical seeded call streams**, so comparisons are paired and the arrival process cancels rather than drowning the policy effect. All reported differences use a paired t-test *and* a Wilcoxon signed-rank test over 30 replications of 7 simulated days, and are called significant only when both agree.

6. Algorithm

The agent is **neural fitted value iteration** over post-decision states — the DQN family, with the max-over-actions taken through the post-decision representation. Because the post-decision state is deterministic given (s, a), $max_a\, Q(s,a) = max_a\, V(s^a)$, so a single value head suffices and no per-station output is baked in — the same agent transfers to a city with a different number of stations.

Training alternates ε-greedy data collection under the current policy with regression onto TD targets $r + \gamma\, max_{a'} V_{target}(s'^{a'})$, computed from a target network frozen once per round. The network is a 2×64 MLP (4,993 parameters) trained with Adam and weight decay. Collection stores every candidate action considered, not merely the one taken, since the target needs a max over actions the behaviour policy declined.

Two further value-based RL methods were implemented for comparison: **LSTD** and **LSPI** over the same features. A cross-entropy policy search over the identical linear class was added as a control: it shares the hypothesis class

exactly, so any gap between it and LSTD is attributable to the training objective rather than to the representation.

Table 1. Headline results (30 paired reps × 7 days).

Policy	Cov.	Mean	Worst zone
Random repositioning	0.357	10.09	0.282
Nearest-idle, static base	0.765	5.82	0.727
MEXCLP stations	0.860	5.38	0.798
DMEXCLP (strong baseline)	0.872	5.26	0.822
Linear ADP, binary reward	0.809	5.78	0.573
Deep NFQ, binary reward	0.724	6.60	0.384
RL agent (NFQ, shaped)	**0.874**	**5.24**	0.818

7. Baseline Comparisons

Four baselines of increasing strength: random repositioning (a floor); nearest-idle dispatch with static home stations, which is what most real services do; stations optimised offline by p-median and MEXCLP; and **DMEXCLP**, the dynamic redeployment heuristic from the EMS literature, which sends a freed unit to whichever station most increases expected covered demand. DMEXCLP is the hard target: greedy action selection on the coverage feature is algebraically identical to it, so the RL agent's hypothesis class provably *contains* the heuristic it is measured against. Losing to it is therefore a statement about the training signal, not about the features.

8. Results

The agent is trained for 8 rounds; the checkpoint is chosen on a validation block of episodes disjoint from both training and test seeds, and all figures below are on the untouched test block (30 paired replications of 7 days).

The trained agent with the shaped reward **reaches parity with DMEXCLP**: coverage 0.874 against 0.872, a difference of +0.002 that is not significant ($p = 0.32$), with mean response, p90 and worst-zone coverage likewise indistinguishable. Against the untrained floor it is +0.052 coverage ($p = 2\times10^{-17}$), so the learning is doing real work — it simply lands where a strong domain heuristic already is. With the binary reward the same architecture collapses to 0.724, and worst-zone coverage falls to 0.384: maximising a count of indicators is maximised by serving dense zones and abandoning sparse ones, an equity failure the pooled metric hides entirely.

The learning curve is flat. Validation coverage oscillates within a band of 0.019 across eight rounds with no upward trend, so additional training did not make the policy better — the agent reaches heuristic parity early and stays there. This is what the project's wider results predict: value-prediction error is dominated by exogenous call variation, so the directions an action can move the state contribute almost nothing to the objective being minimised, and the fit has little incentive to rank actions correctly. A cross-entropy control confirms the reading — optimising policy quality directly over the *same* linear class closes a gap that fitting the value function does not.

Generalisation is partial, and this is the agent's clearest weakness. On a demand profile with relocated hotspots, never seen in training, the agent holds its absolute performance (0.873) and stays +0.063 above the untrained floor, so it has not merely memorised one demand map. But the heuristic *improves* on that easier profile to 0.887, and the agent does not follow it: parity becomes a significant deficit of −0.014 coverage ($p = 4\times10^{-7}$) and −0.033 worst-zone coverage. The learned value function is tuned to the demand distribution it saw; the heuristic recomputes coverage from the live demand model and adapts for free.

How much was ever available. Choosing well is worth +0.091 coverage over random parking, but the best and second-best stations differ by only about 10% of a single marginal gain. Sweeping load (4–6 calls/h) and station count (4–8) leaves this unchanged, so the ceiling is a property of the problem, not of the configuration tested.

9. Conclusions

Four findings, each measured rather than asserted. **(i)** Reward design dominated model class: binary versus shaped reward moved coverage by 0.148, more than any architectural choice tested. **(ii)** Value-prediction accuracy is not policy quality; a control sharing the hypothesis class exactly, but optimising the policy directly, performed better than the fitted value function. **(iii)** The learned policy reaches parity with a strong domain heuristic on the distribution it trained on, but gives that parity up under demand shift, where the heuristic's live recomputation of coverage adapts for free. **(iv)** Reporting that honestly — with confidence intervals, disjoint validation and test seeds, and two agreeing significance tests — is more useful than a tuned victory.

The repository contains the full environment, all agents, the trained network, 70 tests, and scripts reproducing every figure and table here.

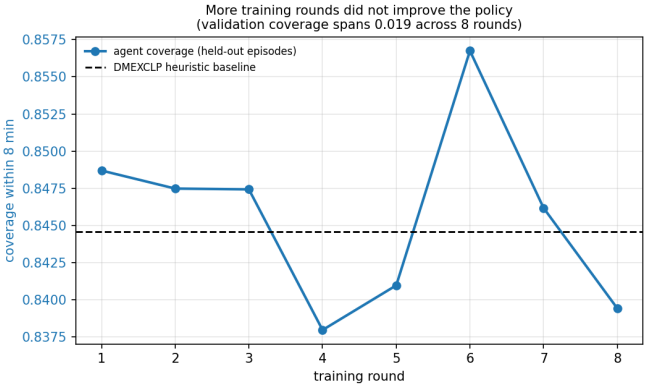


Fig. 1. Validation coverage over training rounds, against the heuristic baseline on the same episodes. Eight rounds produce no upward trend.

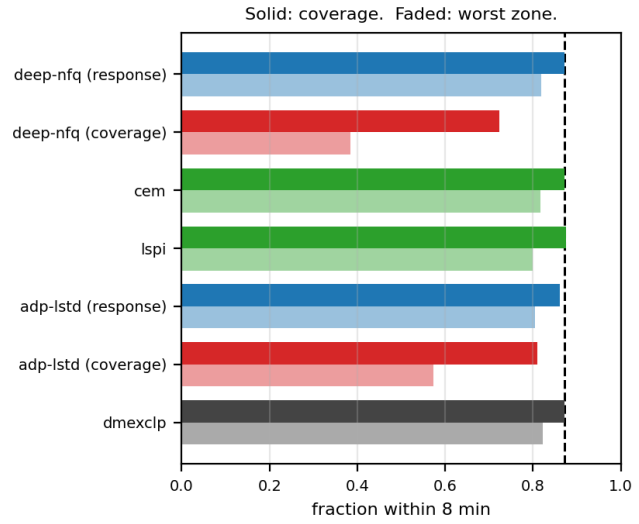


Fig. 2. Reward design dominates model class; the binary reward also collapses worst-zone coverage.